FULL PAPER

Procedural Content Generation in Digital Narratives

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Abstract

This paper demonstrates the Ludus Academik journal template within the Inkwell extension. We present computational examples that generate figures directly from markdown, combining mathematical analysis with reproducible output. The two-column layout, themed headers, and bibliography are all produced from YAML frontmatter and Pandoc compilation.

Keywords: literate programming; reproducible research; Pandoc; LaTeX

1. Introduction

Academic publishing requires precise formatting that varies by journal. Inkwell addresses this by compiling markdown to journal-specific LaTeX classes through Pandoc (MacFarlane 2023). This document uses the Ludus Academik template, producing a two-column layout with themed section headers.

The literate programming paradigm (Knuth 1984) allows code and prose to coexist. Inkwell extends this to compiled PDF output: code blocks execute, and their results (figures, tables, text) appear in the final document.

2. Computational Example

We demonstrate with a Fourier series visualization (**?**). The partial sum approximating a square wave is given by equation 1.

$$f_n(x) = \sum_{k=1}^{n} \frac{4}{(2k-1)\pi} \sin\big((2k-1)x\big) \tag{1}$$

Figure **??** shows the partial sums converging to the square wave as n increases. The overshoot at the discontinuity is the Gibbs phenomenon.

```
{python file=".inkwell/scripts/sine_plot.py"
output="sine_plot" caption="Fourier
partial sums converging to a square wave."
label="fourier"}
```

3. Data Visualization

Figure **??** shows a simulated scatter plot with linear regression, generated inline by Python.

```
{python file=".inkwell/scripts/scatter.py"
output="scatter" caption="Simulated regression
with n = 150 data points." label="scatter"}
```

4. Results

Table **??** shows the convergence behavior of the Fourier partial sums at the midpoint $x = \pi/2$, where the true value is $f(x) = 1$. The peak overshoot column quantifies the Gibbs phenomenon: regardless of n, the maximum value overshoots by approximately 9% of the jump magnitude.

```
{python file=".inkwell/scripts/convergence_table.py"
output="convergence" caption="Convergence
of Fourier partial sums at x = pi/2."
label="convergence"}
```

5. Conclusion

The regression in Figure **??** was fitted to $n = sample_n$ observations, yielding $r = corr_r$ and $\hat{\beta} = `pythonf"float(slope):.2f"`$. As shown in Figure **??** and Figure **??**, Inkwell produces publication-quality figures from Python scripts. Table **??** in section 4 demonstrates CSV-to-table rendering, and Equation 1 in section 2 confirms that LaTeX math compiles correctly. The Ludus template handles all of these in a two-column layout with themed headers and bibliography.

References

Knuth, Donald E. 1984. "Literate Programming." *The Computer Journal* 27 (2): 97–111. https://doi.org/10.1093/comjnl/27.2.97.

MacFarlane, John. 2023. *Pandoc: A Universal Document Converter*. Released. https://pandoc.org.

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